

# Geometry in the entrance paper, and the annual review

What the examination actually asks, and what two years of geometry come to.

LESSON 67–68

2 × 40 MIN

GEOMETRY 11, YAKUNIY TAKRORLASH

EXAMINATION TECHNIQUE

LESSON CLOCK

15'

30'

20'

20'

15'

|                           |        |
|---------------------------|--------|
| The seven types           | 15 min |
| A timed mixed set         | 30 min |
| Diagnosis by type         | 20 min |
| The two years in one page | 20 min |
| The plan                  | 15 min |

LEARNING OBJECTIVES

- Recognise the geometry question types of the entrance paper.
- Choose the fastest of the synthetic, coordinate and vector methods.
- Work a mixed timed set and diagnose the errors by type.
- Leave the course with a usable summary and a plan.

TEXTBOOK

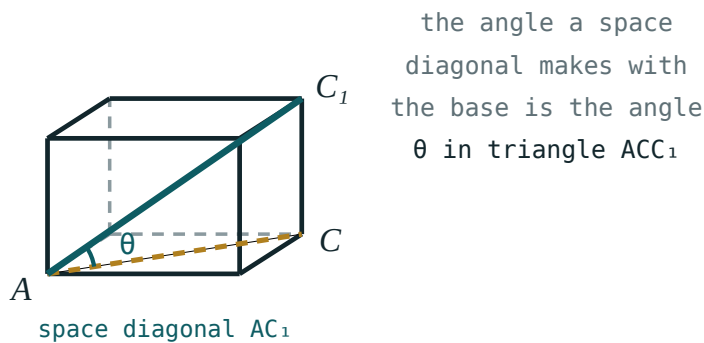
|           |                              |
|-----------|------------------------------|
| National  | pp. 345–352                  |
| Cambridge | Core & Extended, pp. 405–408 |
| Workbook  | Revision set G5              |

01

## Explanation

### The seven types

| TYPE                  | TYPICAL QUESTION                       | FASTEST METHOD                        |
|-----------------------|----------------------------------------|---------------------------------------|
| plane figure          | a triangle's area or an angle          | synthetic, with a sketch              |
| circle                | a tangent, a chord, an inscribed angle | coordinates if an equation is given   |
| coordinate            | a line, a circle, an intersection      | algebra straight through              |
| solid:<br>measurement | a volume or an area                    | the nine formulas                     |
| solid: angle          | a line and a plane, or a dihedral      | vectors in a box, synthetic elsewhere |
| solid: section        | name and measure a cut                 | synthetic, three rules                |
| similarity            | a ratio of lengths, areas or volumes   | $k, k^2, k^3$                         |



One box, and five of the seven types measured inside it.

THE FIRST LINE OF EVERY ANSWER

Name the solid or the figure, and write down what you know:  $B$ ,  $h$ ,  $r$ , the coordinates, the vectors. The question rarely names them for you, and the mark scheme rarely gives anything before that line.

A timed mixed set

Twelve questions, twenty minutes. Mark anything that takes over ninety seconds and come back to it.

| #  | QUESTION                                        | ANSWER                            |
|----|-------------------------------------------------|-----------------------------------|
| 1  | Area of a triangle with sides 13, 14, 15        | 84                                |
| 2  | Tangent to $x^2 + y^2 = 25$ at (3, 4)           | $3x + 4y = 25$                    |
| 3  | Distance from (1, 8) to $y = 2x$                | $\frac{6}{\sqrt{5}} \approx 2.68$ |
| 4  | Volume of a cone, $r = 5$ , $h = 12$            | $100\pi$                          |
| 5  | Curved area of that cone                        | $65\pi$                           |
| 6  | Space diagonal of a $3 \times 4 \times 12$ box  | 13                                |
| 7  | Angle of that diagonal with the base            | $\approx 67.4^\circ$              |
| 8  | Section of a cube through six edge-midpoints    | regular hexagon                   |
| 9  | Volume of a sphere of surface area $144\pi$     | $288\pi$                          |
| 10 | Two similar solids, volumes 27 : 64: area ratio | 9 : 16                            |
| 11 | Angle between (1,2,2) and (2,2,1)               | $\approx 27.3^\circ$              |
| 12 | Least-metal closed cylinder: $\frac{h}{r}$      | 2                                 |

## Diagnosis by type

Do not count the score. Sort the errors.

| ERROR TYPE        | LOOKS LIKE                               | WHAT TO PRACTISE                 |
|-------------------|------------------------------------------|----------------------------------|
| <b>reading</b>    | curved when total was asked              | the three questions, written out |
| <b>formula</b>    | a missing $\frac{1}{3}$ or $\frac{4}{3}$ | the nine-formula page, daily     |
| <b>method</b>     | the long route to a short answer         | the method table above           |
| <b>arithmetic</b> | a right method, a wrong number           | slow down on the last line       |
| <b>units</b>      | cm <sup>3</sup> where litres were wanted | convert lengths first            |

## THE NUMBER THAT PREDICTS THE EXAM SCORE

Not the practice score — the number of errors of the **same type** repeated across two papers. One repeated type is worth four or five marks in the real examination, and each one takes about twenty minutes of targeted practice to remove.

## The two years in one page, and the plan

| GRADE 10                                        | GRADE 11                                  |
|-------------------------------------------------|-------------------------------------------|
| axioms; three points determine a plane          | coordinates; a point is three numbers     |
| parallelism: parallel to a line <b>in</b> it    | vectors; parallel means a scalar multiple |
| perpendicularity: two <b>intersecting</b> lines | the scalar product; $u \cdot v = 0$       |
| angles by projection                            | angles by dot product                     |
| sections of polyhedra                           | prisms, pyramids, cones, spheres          |
| the plane coordinate method                     | mensuration in context, and optimisation  |

## ONE SENTENCE FOR THE WHOLE OF SCHOOL GEOMETRY

Every problem is solved by finding the one right triangle it contains — or, with vectors, the one dot product it depends on. Two years, one idea: reduce what is in front of you to something you can already measure.

**The plan.** Between now and the examination: the nine formulas rewritten from memory once a week; one timed mixed set a week, sorted by error type; and the list of repeated types kept on a single sheet. Not more mathematics — the same mathematics, faster and with fewer repeated slips.

That is the end of the geometry course. The formulas will fade; the habit of drawing the figure before writing anything will not, and it is the more useful of the two.

## 02 Worked examples

**EXAMPLE 1** Model answer, №1: the area of a triangle with sides 13, 14, 15.

①  $s = 21$

Heron.

②  $S = \sqrt{21 \times 8 \times 7 \times 6}$

③  $= \sqrt{7056} = 84$

A standard triangle worth remembering.

ANSWER

84

**EXAMPLE 2** Model answer, №7: the angle of the space diagonal of a  $3 \times 4 \times 12$  box with the base.

① Base diagonal  $\sqrt{9 + 16} = 5$ .

② Space diagonal  $\sqrt{25 + 144} = 13$ .

③  $\tan \theta = \frac{12}{5}$

④  $\theta \approx 67.4^\circ$

ANSWER

$\approx 67.4^\circ$

**EXAMPLE 3** Model answer, №11: the angle between  $(1,2,2)$  and  $(2,2,1)$ .

$$1 \quad u \cdot v = 2 + 4 + 2 = 8$$

$$2 \quad |u| = |v| = 3$$

$$3 \quad \cos \theta = \frac{8}{9}$$

$$4 \quad \theta \approx 27.3^\circ$$

**ANSWER**

$$\approx 27.3^\circ$$

**EXAMPLE 4** Model answer, №9: the volume of a sphere of surface area  $144\pi$ .

$$1 \quad 4\pi R^2 = 144\pi \Rightarrow R = 6$$

$$2 \quad V = \frac{4}{3}\pi(216)$$

$$3 \quad = 288\pi \approx 905$$

Not  $144\pi$  — area and volume are different powers.

**ANSWER**

$$288\pi$$

**03** Interactive model

Run the twelve questions strictly to time, then spend the rest of the lesson only on the error-type table.

**The two years in twelve questions**

A plane is determined by:

two points

three non-collinear points

one line

four points

Two lines that never meet are:

parallel

parallel or skew

skew

intersecting

A line  $\perp$  to a plane is  $\perp$  to:

one line

two intersecting lines, then all

two parallel lines

the normal

The angle to a plane is measured to:

the normal

the projection

any line

the edge

Volume of a prism:

$Ph$

$Bh$

$\frac{1}{3} Bh$

$2Bh$

Volume of a cone:

$\pi r^2 h$

$\frac{1}{3} \pi r^2 h$

$\pi r l$

$\frac{1}{3} \pi r^2 l$



Curved area of a cone:

$$\pi r^2$$

$$\pi r l$$

$$\pi r h$$

$$2\pi r l$$

Surface area of a sphere:

$$4\pi R^2$$

$$\frac{4}{3}\pi R^3$$

$$2\pi R^2$$

$$\pi R^2$$

Volumes 27 : 64 give areas in:

$$27 : 64$$

$$9 : 16$$

$$3 : 4$$

$$81 : 256$$

A section of a cube has at most:





$u \perp v$  when:

 $u \cdot v = 1$ 

 $u \cdot v = 0$ 

 $u = v$ 

 $|u| = |v|$ 

Least-metal closed cylinder:

 $h = r$ 

 $h = 2r$ 

 $h = \frac{r}{2}$ 

 $h = 3r$ 

## 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH              | O'ZBEKCHA           | РУССКИЙ               |
|----------------------|---------------------|-----------------------|
| Entrance examination | Kirish imtihoni     | Вступительный экзамен |
| Question type        | Savol turi          | Тип задачи            |
| Synthetic method     | Sintetik usul       | Синтетический метод   |
| Coordinate method    | Koordinatalar usuli | Координатный метод    |
| Vector method        | Vektor usuli        | Векторный метод       |
| Time management      | Vaqtni taqsimlash   | Распределение времени |
| Error type           | Xato turi           | Тип ошибки            |
| Summary sheet        | Xulosa varaqasi     | Итоговый лист         |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

The first line of an entrance answer should be:

the formula

the named figure and what is known

the answer

a guess

For an angle inside a box, prefer:

synthetic

vectors

estimation

trial

After a practice paper, count:

the score

the repeated error types

the time

the blanks

The one idea of two years of geometry:

memorise formulas

reduce to a right triangle or a dot product

draw neatly

work fast

## 06 Practice · 21 problems

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Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Area of a triangle with sides  $13, 14, 15$

ANSWER  $84$

2. Tangent to  $x^2 + y^2 = 25$  at  $(3, 4)$

ANSWER  $3x + 4y = 25$

3. Volume of a cone,  $r = 5, h = 12$

ANSWER  $100\pi$

4. Curved area of that cone

ANSWER  $65\pi$

5. Space diagonal of a  $3 \times 4 \times 12$  box

ANSWER  $13$

6. Radius of a sphere of area  $144\pi$

ANSWER  $6$

7. Volumes  $27 : 64$ : area ratio

ANSWER  $9 : 16$

Medium · the standard question

7 PROBLEMS

1. Angle of the  $3 \times 4 \times 12$  space diagonal with the base

ANSWER  $\approx 67.4^\circ$

2. Distance from  $(1, 8)$  to  $y = 2x$

ANSWER  $\frac{6}{\sqrt{5}} \approx 2.68$

3. Section of a cube through six edge-midpoints

ANSWER A regular hexagon

4. Volume of a sphere of area  $144\pi$

ANSWER  $288\pi$

5. Angle between  $(1,2,2)$  and  $(2,2,1)$

ANSWER  $\approx 27.3^\circ$

6. Least-metal closed cylinder:  $\frac{h}{r}$

ANSWER 2

7. Total area of a cone,  $r = 5$ ,  $h = 12$

ANSWER  $90\pi$

Hard · stretch and reason

7 PROBLEMS

1. A cube of edge  $a$ : the distance from a vertex to the plane of the three adjacent vertices

ANSWER  $\frac{a\sqrt{3}}{3}$

2. A regular tetrahedron of edge  $a$ : its volume

ANSWER  $\frac{a^3\sqrt{2}}{12}$

3. A sphere inscribed in a cone of radius 6 and height 8

ANSWER  $r = 3$

4. A cone of radius 9 cut halfway up: the frustum as a fraction of the cone

ANSWER  $\frac{7}{8}$

5. The regular hexagonal section of a cube of edge  $a$ : its area

ANSWER  $\frac{3a^2\sqrt{3}}{4}$

6. Two spheres of radii 3 and 5 touch externally: the distance between the centres

ANSWER 8

7. A closed cylinder of volume  $V$  with least area: express that area in terms of  $V$

ANSWER  $3\sqrt[3]{2\pi V^2}$

## 07 Homework

### Homework — 4 tasks

This is the last geometry lesson of the school course. The summary page is the thing to keep.

1. Work the twelve-question set again under timed conditions and sort every error by type.
2. Write the two years in one page: six lines for Grade 10, six for Grade 11, with a sketch for each.
3. Write the nine formulas from memory, check them, and note which you got wrong.



4. Write your revision plan for the weeks before the examination in three sentences, and date it.